

Example 1:

Write a program in VB. Net to read array B(3, 3), then summation of all elements in this array using For .. Next Statement?

Writing the Code

Use the Code Editor

- Double-click the Sum Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click_1(sender As Object, e As EventArgs) Handles Button1.Click
    Dim I, J As Integer
    Dim B(3, 3) As Integer
    Dim Sum As Integer = 0
    For I = 1 To 3
        For J = 1 To 3
            B(I, J) = InputBox("Enter The Values of Array B ")

            Next J
        Next I
    For I = 1 To 3
        For J = 1 To 3
            Sum = Sum + B(I, J)

            Next J
        Next I
    For I = 1 To 3
        For J = 1 To 3
            Console.Write(B(I, J) & " ")

            Next J
        Console.WriteLine()
    Next I
    Console.WriteLine()
    MessageBox.Show("The Summation is " & Sum)

End Sub
```

Example 2:

Design the following form and write a program in VB. Net to read two arrays A(3,3), B(3,3), then find the summation of elements between arrays A and B in new array C(3,3) using For .. Next Statement?

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Sum
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	Array A
Labele2	Text	Array B
Labele3	Text	Array C
Form1	Text	Summation of Two Arrays
Form1	MaximizeBox	False
Form1	MinimizeBox	False

ListBox1	Items.add()	Elements of A(3, 3)
ListBox2	Items.add()	Elements of B(3, 3)
ListBox3	Items.add()	Elements of C(3, 3)

Writing the Code

Use the Code Editor

1. Double-click the Sum Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click_1(sender As Object, e As EventArgs) Handles Button1.Click
```

```
    Dim I, J As Integer
```

```
    Dim A(3,3) As Integer
```

```
    Dim B(3,3) As Integer
```

```
    Dim C(3,3) As Integer
```

```
    For I = 1 To 3
```

```
        For J=1 To 3
```

```
            A(I, J) = InputBox("Enter The Elements of Array A")
```

```
            ListBox1.Items.Add(A(I, J))
```

```
        Next J
```

```
    Next I
```

```
    For I = 1 To 3
```

```
        For J=1 To 3
```

```
            B(I, J) = InputBox("Enter The Elements of Array B")
```

```
            ListBox1.Items.Add(B(I, J))
```

```
        Next J
```

```
    Next I
```

```
For I = 1 To 3
    For J=1 To 3
        C(I, J) = A(I, J) + B(I, J)
        ListBox3.Items.Add(C(I, J))
    Next J
Next I
End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ListBox1.Items.Clear()
    ListBox2.Items.Clear()
    ListBox3.Items.Clear()
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Me.Close()
End Sub
```

Example 3:

Write a program in VB. Net to read array A(3,3), then find the summation of all elements in the main diagonal array using For .. Next Statement?

Writing the Code

Use the Code Editor

- Double-click the Sum Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
```

```
    Dim I, J As Integer
```

```
    Dim A(3, 3) As Integer
```

```
    Dim Sum As Integer = 0
```

```
    For I = 1 To 3
```

```
        For J = 1 To 3
```

```
            A(I, J) = InputBox("Enter The Values of Array A ")
```

```
        Next
```

```
    Next
```

```
    For I = 1 To 3
```

```
        For J = 1 To 3
```

```
            Console.Write(A(I, J) & " ")
```

```
        Next
```

```
        Console.WriteLine()
```

```
    Next
```

```
    For I = 1 To 3
```

```
        For J = 1 To 3
```

```
            If I = J Then
```

```
                Sum = Sum + A(I, J)
```

```
            End If
```

```
        Next
```

Next

```
Console.WriteLine()  
    MessageBox.Show("The Summation is " & Sum)
```

End Sub

Example 4:

Design the following form and write a program in VB. Net to read array A(3,3), then find the account of Odd and Even elements in this array using For .. Next Statement?

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Find
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	Odd Number

Labele2	Text	Even Number
Labele3	Text	Result of Odd Number
Labele 4	Text	Result Of Even Number
ListBox1	Items.add()	Odd Number
ListBox2	Items.add()	Even Number

Writing the Code

Use the Code Editor

1. Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs)
Handles Button1.Click
    Dim I, J As Integer
    Dim A(3, 3) As Integer
    Dim C1 As Integer = 0
    Dim C2 As Integer = 0

    For I = 1 To 3
        For J = 1 To 3
            A(I, J) = InputBox("Enter The Values of Array A ")
        Next J
    Next I

    For I = 1 To 3
        For J = 1 To 3
            If A(I, J) Mod 2 = 0 Then
                ListBox2.Items.Add(A(I, J))
                C2 = C2 + 1
                Label4.Text = C2
            Else
                ListBox1.Items.Add(A(I, J))
                C1 = C1 + 1
                Label3.Text = C1
            End If
        Next J
    Next I
End Sub
```

```
        Next J
    Next I
    For I = 1 To 3
        For J = 1 To 3
            Console.Write(A(I, J) & " ")
        Next J
        Console.WriteLine()
    Next I

End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ListBox1.Items.Clear()
    ListBox2.Items.Clear()
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Me.Close()
End Sub
```